

MAHMUD GHNAJ

Senior Software Engineer | Full-Stack Engineer

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 6+ years of experience building high-performance, scalable applications, with deep specialization in AI-powered platforms, healthcare systems, and large-scale IoT solutions. Proven track record leading frontend teams, architecting micro-frontend systems, and delivering real-time data platforms that improve performance, reduce costs, and scale to thousands of concurrent assets.

PROFESSIONAL EXPERIENCE

Senior Frontend Developer | Ntira AI Applications and Services

Jan 2025 – Present

- Developed front-end and backend solutions for complex, AI-powered healthcare platforms supporting multiple user roles (doctors, nurses, and other medical staff) using Next.js and Node.js.
- Built and integrated AI-powered search and assistant tools, similar to Claude and ChatGPT, customized for the medical domain.
- Developed data integration pipelines to extract information from legacy healthcare systems and feed it into backend services for processing and AI-driven automation.
- Migrated a legacy React application to a scalable micro-frontend architecture, improving maintainability and enabling independent feature development.
- Designed and implemented the front-end architecture of an AI-enabled hospital management platform supporting complex medical workflows.
- Developed AI-powered browser automation tools as Chrome Extensions to streamline operations across enterprise medical platforms.
- Applied Domain-Driven Design (DDD) principles and factory patterns to structure scalable, maintainable AI-driven system architectures.

Frontend Lead | Unifi Solutions

Jul 2024 – Dec 2024

- Managed and led a team of 4 frontend developers, overseeing the full development lifecycle.
- Increased team efficiency by 20% through effective task delegation.
- Optimized performance, reducing load times from 8s to under 1s, boosting user retention by 30%.
- Developed a reusable frontend component library, cutting development time by 40%.
- Built real-time data visualization tools using D3.js and Highcharts, improving decision-making.
- Designed a high-performance tracking system monitoring 5,000+ vehicles, reducing operational costs by 15%.
- Led platform localization efforts, ensuring seamless multilingual support.

Senior Frontend Developer | Unifi Solutions

Apr 2023 – Jun 2024

- Established the foundation of a scalable component library, laying the groundwork for consistent UI across multiple projects.
- Improved application performance and responsiveness through targeted frontend optimizations, reducing page load times.
- Developed dynamic form builders and configurable UI components, enhancing platform adaptability across use cases.
- Combined WebSockets and GraphQL to power real-time data streaming into interactive, live dashboards built with D3.js, Recharts, and Chart.js.
- Refactored legacy codebases to align with modern frontend best practices, improving code quality and long-term maintainability.
- Implemented state management solutions with Zustand & Redux, improving scalability across growing application complexity.

Full Stack Developer | Disrupt-X

May 2022 – Jun 2023

- Led the migration of core backend services from Express.js to NestJS, improving code structure, scalability, and long-term maintainability.
- Developed and maintained backend services using Node.js, NestJS, and MongoDB to support scalable IoT platform features.
- Implemented authentication and authorization systems using secure token-based mechanisms to protect platform APIs.
- Designed and integrated RESTful APIs to support real-time data processing and communication between services.
- Collaborated on full-stack features connecting frontend dashboards with backend data services.
- Optimized database queries and backend performance to support large-scale IoT data ingestion and processing.

Frontend Engineer | Disrupt-X

May 2021 – May 2022

- Developed large-scale, complex IoT platforms using React and Vue, featuring intricate map views, data charts, and multi-layered dashboard interfaces.
- Built a role-based access control system supporting multiple user roles and permission levels across the platform.
- Engineered a real-time vehicle tracking system for 5,000+ assets, cutting operational costs by 15% through performance and architecture improvements.
- Designed and built a dynamic reporting engine using Vue.js, giving users fully customizable analytics dashboards.
- Led end-to-end Google Maps integration, implementing live tracking, geofencing, and marker clustering for large-scale map data.
- Built multilingual platform support, expanding the product's reach to global users.
- Developed a reusable UI component library, improving modularity and accelerating future feature development.

Full-Stack Developer | Freelance

Jan 2020 – May 2021

- Took full ownership of two freelance projects end-to-end — a real estate management platform and a car dealership platform — covering frontend, backend, and deployment.
- Built both platforms using Next.js, React, Node.js, and MySQL, handling architecture decisions from scratch.
- Deployed and maintained both applications in production, managing hosting and release processes independently.
- Implemented social media authentication (Google, Facebook, Apple) to simplify and secure user sign-in.
- Optimized SEO and performance using server-side rendering (SSR) with Next.js, improving search visibility and load times.
- Optimized systems to handle high data volumes smoothly.

TECHNICAL SKILLS

Frontend: Next.js, React.js, TypeScript, JavaScript, Zustand, Redux, Context API, Tailwind CSS, styled-components, Material UI, Bootstrap, CSS3, HTML5, Web3

Backend: Node.js, NestJS, REST APIs

Data & Performance: WebSockets, GraphQL, MongoDB, MySQL, D3.js, Highcharts, Chart.js

Cloud & DevOps: Docker, CI/CD (GitHub Actions, Jenkins, Bitbucket), AWS, Firebase

Other: Blockchain (Ethereum, Solidity, Web3.js), OAuth, Localization, PWA

EDUCATION

B.Sc. in Computer Science | Al-Baath University

2016 – 2020

Thesis: Developed a secure electronic voting system using Ethereum, Solidity, Truffle, and Web3.js.

KEY PROJECTS

- [Disrupt-X](#) — Enterprise IoT & AI analytics platform
- [ABIOT](#) — IoT marketplace for smart city solutions
- [climacert X](#) — Smart building automation and facility management
- [Medxsol](#) — AI-enabled healthcare medication management system
- [ALEF](#) — Multi-protocol IoT device management platform
- [Fleet Management](#) — GPS fleet platform for 5,000+ vehicles
- [Asset Watch](#) — Indoor IoT asset tracking system
- [Cracking IoT](#) — The largest IoT event in MENA
- [Asset Management](#) — AI-powered facility monitoring platform
- [IBMS](#) — Full building automation and control system
- [Aligner-KDS](#) — Kitchen display order system
- [Ignite Shield](#) — IoT facility and asset monitoring solution